Module 21: Power BI Integration

Contents

Module 21: Power BI Integration 1
Prerequisites: 2
Scenario 3
Practice: Power BI – Getting Started 3
Practice: Adding Power BI tab in Workspace 4
Practice: Power BI on Dynamics 365 for Ops Workspace 4

Prerequisites:

1. To complete this lab you will be using the VM provided to you.
    a. Login: builtin\AXLocalAdmin
    b. Password: <Available in LCS>
2. Open Visual Studio as administrator
    a. Right Click Visual Studio icon, Run as administrator

Scenario

For Asset Insurance System scenario, create a Power BI Visuals pulling data from Dynamics 365 for Finance & Operations. Publish the Visuals on the Power BI dashboard. Publish the same visuals in the custom Workspace developed in Dynamics 365 for Finance & Operations.

Practice: Power BI – Getting Started

1. Download Power BI Desktop from Microsoft Site – select 32 bit or 64 bit (as per your machine & office 365 configuration)
2. Install Power BI Desktop and open the application
3. In the top right corner, there is an option to sign in. Click on it and sign in using your LCS credential
4. Click Get Data -> OData Feed
5. Enter your AX7 url (after '/' enter 'data')
6. Select AssetInsurancePremiums from the list of Data Entities in the Navigator window
7. In the left side of the Power BI workspace, you will find three tabs – Report (Currently Selected), Data & Relationships
8. Please select Data
9. Click on New Column button under the "Modeling" action pane
10. A new column will be created in the Data sheet, with placeholder to write value of the newly created calculated field
11. Please type following value in the calculated field

```
TotalPremium = AssetInsurancePremiums[PolicyAmount] *
SWITCH(AssetInsurancePremiums[AISPaymentCycle], "Monthly", 12, "Quarterly", 4,
"Halfyearly", 2, "Yearly", 1, "None", 0)
```

12. Go back to the Report tab
13. Select Clustered Column chart from the Visualization list
14. Drag AssetGroup field under Axis
15. Drag InsuranceVendor field under Legend
16. Drag TotalPremium under Values; change the aggregate to sum
17. Save the visual as a pbix file
18. Click on the Publish button at the top right end

19. Open https://powerbi.microsoft.com/ and sign in with your LCS credential
20. Under My Workspace -> Reports, you should find your new visuals
21. Add that in a new dashboard by clicking on pin (third button from top right of the visuals)

Practice: Adding Power BI tab in Workspace

1. Open Form AISAssetInsuranceWorkspace from Solution Explorer
2. Add a new tabpage under Panorama Tab in the Design
3. Select "Section Power BI" as the pattern of the tabpage
4. Add "Power BI Control" under the new tabpage
5. In the Solution Explorer, right click the Project 'Asset Insurance System'
6. Click Build Solution (or use Ctrl+Shift+B)
7. Review the Output screen to review process

Practice: Power BI on Dynamics 365 for Ops Workspace

1. Open https://dev.powerbi.com/apps
    a. Login using your LCS credential
    b. Type "Dynamics 365 for Ops" as App Name
    c. Type Dynamics 365 Ops URL suffixed by "oauth" in the Redirect URL field
    d. Type https://app.powerbi.com/ in the Homepage URL field
    e. Choose the APIs to access and click on register App
    f. Client ID and Client Secret field will be populated with some random numbers; store them in a notepad file
2. Go to Dynamics AX -> System administration > Setup > Power BI to open the Power BI configuration page
3. Specify the following values for the fields on this form
    - Enabled: Set the option to Yes.
    - Azure AD authority URI: The correct value, https://login.windows.net, should already be entered when you open the page.
    - Azure AD Power BI resource URI: The correct value, https://analysis.windows.net/powerbi/api, should already be entered.
    - Azure AD Tenant: The correct value, axvm.onmicrosoft.com, should already be entered.

- Client ID: Enter the Client ID value that you recorded earlier.
- Application key: Enter the Secret Key value that you recorded earlier.
- Redirect URL: The correct value, the root URL of your deployment plus oauth, should already be entered.
- Power BI API Address: https://api.powerbi.com/beta/myorg, should already be entered.
- Apply tile filter: The correct value, Yes, should already be set.
- Tile filter table: The correct value, Entities, should already be entered.
- Tile filter column: The correct value, ID, should already be entered.
- Click Save to save your changes

4. Open Asset Insurance Workspace
5. In the Power BI banner, click **Get started**. If this is the first time that you've started Power BI from Dynamics 365 for Finance & Operations, you're prompted to authorize sign-in to Power BI from the D365 client. Click **Click here to provide authorization to Power BI**.
6. Because you're already signed in to AAD in Dynamics AX, you don't have to enter your credentials again. A new tab appears, where you're prompted to authorize the connection between Dynamics AX and Power BI. You can now return to the original tab.
7. The Add / remove Power BI tiles slider dialog opens. The tabs on the left show your dashboards, and the related visualizations appear on the right. You can now select visualizations to add (pin) to the workspace.
8. After you've finished selecting visualizations, click OK
9. The Power BI visual should be published in your new D365 for Fin & Ops workspace